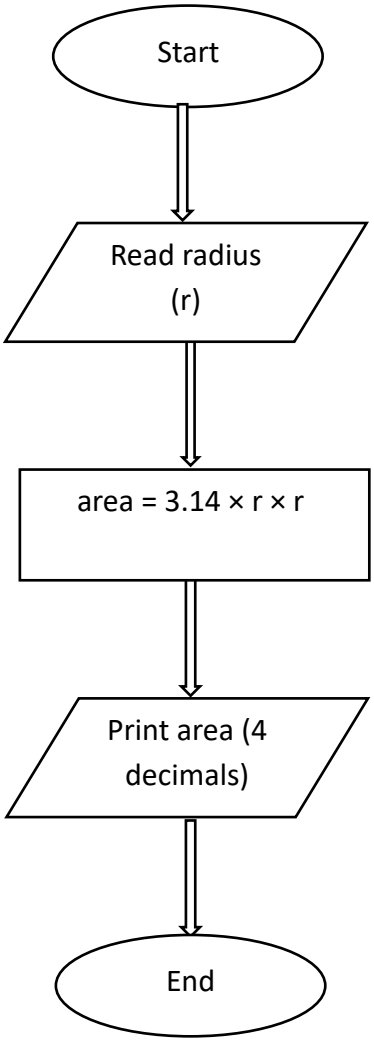
Experiment 1.1

Algorithm:

- Start
- Read the radius `r` (floating-point number)
- Set π = 3.14
- Calculate area using the formula = area= π×r×r
- Display the area
- End

Flowchart:



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1.1.1. Area of Circle

14:02

Write a Python program that calculates the area of a circle when the radius is provided by the user. Use π = 3.14 and display the area.

Input Format:

- A single line containing a floating-point number representing the radius.

Output Format:

- Print the computed area of the circle formatted to 4 decimal places.

Sample Test Cases +

Explorer

circlearea...

Submit

```
radius = float(input())
area = 3.14*radius*radius
print(f"{area:.4f}")
```

Debugger

Average time	Maximum time
0.010 s	0.013 s
9.75 ms	13.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 13 ms

Debug

Expected output	Actual output
3.36	3.36
35.4493	35.4493

Test case 2 11 ms

Terminal Test cases

‹ Prev Reset Submit Next ›

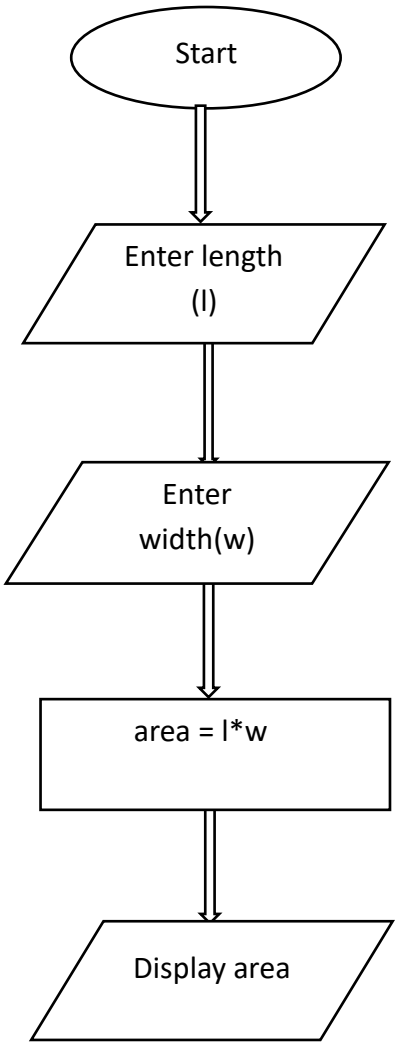
Experiment 1.2

Algorithm:

- Start
- Read the length of the rectangle
- Read the width of the rectangle
- Calculate area using: area= length×width
- Display the area formatted to 2 decimal places
- End

End

Flowchart:



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1.1.2. Area of Rectangle

04:16

Write a Python program to calculate the area of a rectangle given its length and width.

Formula:

$$\text{Area of Rectangle} = \text{Length} \times \text{Width}$$

Input Format:

- First line contains a float value representing the length of the rectangle
- Second line contains a float value representing the width of the rectangle

Output Format:

- Print the area of the rectangle as a float value formatted to 2 decimal places.

Sample Test Cases +

Explorer

areaOfRe...

Submit

```python
# Type Content here...
length = float(input())
width = float(input())
area = length * width
print(f"{area:.2f}")
```

Average time 0.012 s 12.30 ms

Maximum time 0.016 s 16.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1 16 ms

Debug

Expected output	Actual output
10.5	10.5
5.2	5.2
54.60	54.60

Test case 2 15 ms

Terminal Test cases

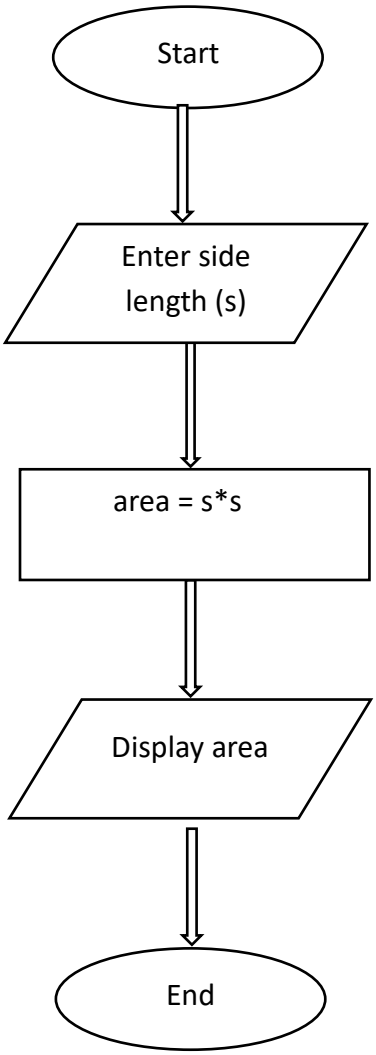
Debugger

Experiment 1.3

Algorithm:

- Start
- Read the side length of the square
- Calculate the area using: area=side×side
- Display the area
- End

Flowchart:



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1.1.3. Calculate Area of the Square

04:14

Write a Python program that prompts the user to enter the *side_length* of a square and computes the area of the square.

Formula:

- $\text{Area} = \text{side_length}^2$

Input Format:

- The input is a positive integer value that represents the *side_length* of the square.

Output Format:

- The output is a positive integer value that represents the area of the square.

Sample Test Cases +

Explorer

AreaSqua...

Submit

```python
# Write your code here...
side_length = int(input())
area = side_length * side_length
print(area)
```

Debugger

Average time	Maximum time
0.009 s	0.013 s
9.00 ms	13.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 13 ms

Debug

Expected output	Actual output
5	5
25	25

Test case 2 6 ms

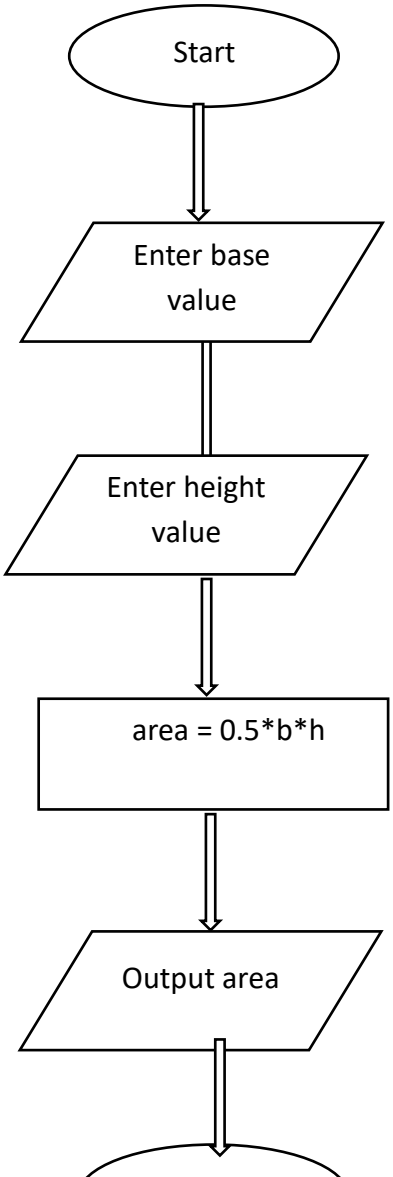
Experiment 1.4

Algorithm:

- Start
- Read the value of base

  Read the value of height
- Calculate area using the formula :area = 0.5 × base × height
- Display the value of area (formatted to two decimals)
- End

Flowchart:



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1.1.4. Area of Triangle

16:11

Write a Python program that prompts the user to enter the triangle's base and height and computes the triangle's area.

Formula: $Area\ of\ Triangle\ =\ 0.5\ \times base\ \times height.$

Input Format:

- The first line of input is the float value that represents the base of the triangle.
- The second line of input is the float value that represents the height of the triangle.

Output Format:

- The output is the floating point value that represents the area of a triangle, formatted to two decimals.

Sample Test Cases +

Explorer

triangleA...

Submit

```
base = float(input())
height = float(input())

area = 0.5 * base * height
print(f"{area:.2f}")
```

Debugger

Average time	Maximum time
0.015 s	0.023 s
15.00 ms	23.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 23 ms

Debug

Expected output	Actual output
6.54	6.54
1.23	1.23
4.02	4.02

Test case 2 8 ms

Terminal Test cases

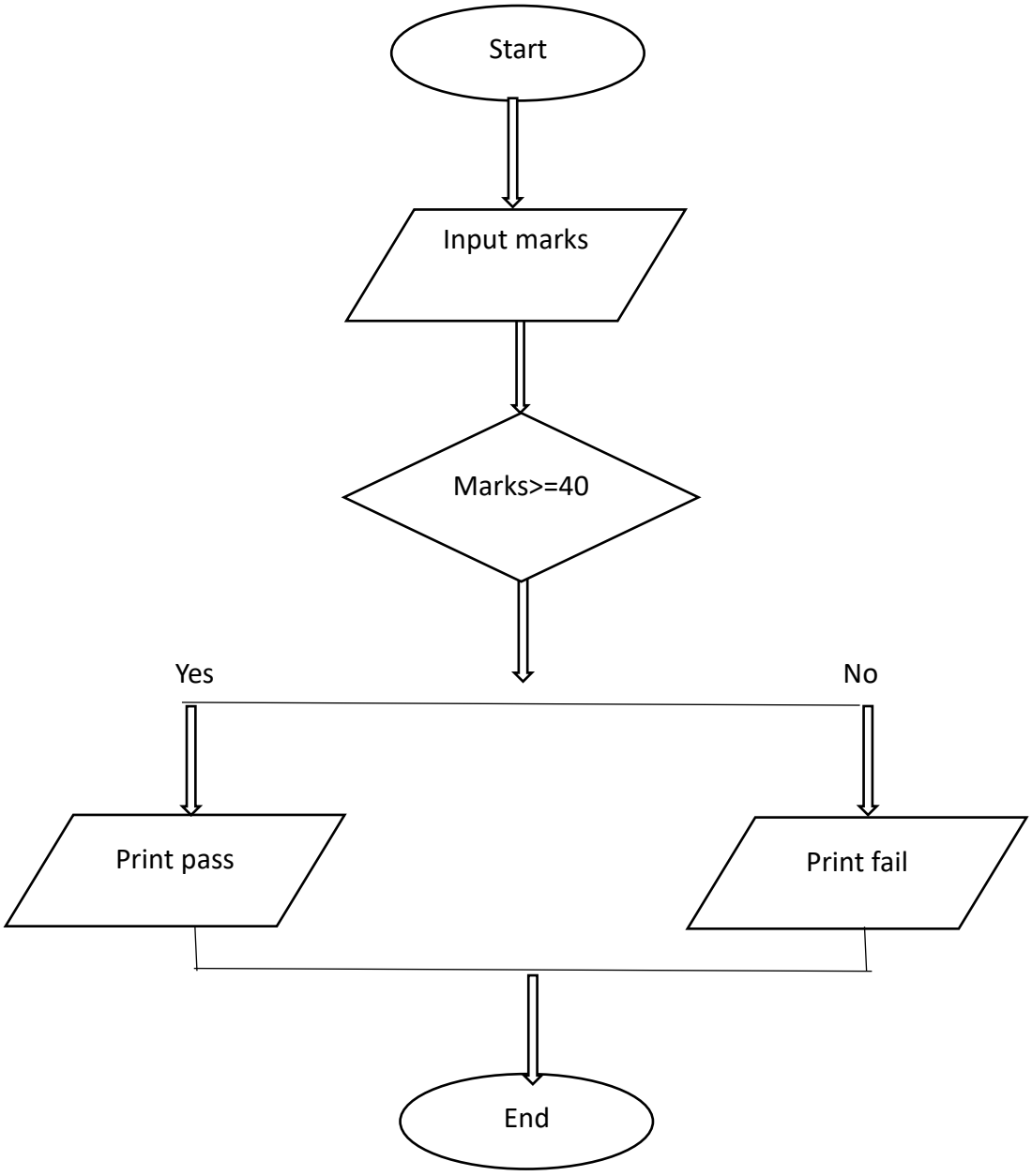
‹ Prev Reset Submit Next ›

Experiment 1.5

Algorithm:

- Start
- Read the value of marks
- Check if marks ≥ 40
  - If yes, print "Pass"
  - If no, print "Fail"
- End

Flowchart:



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1.1.5. Student Pass or Fail Status

08:09

Write a Python program to determine whether a student passed the exam or not based on their marks.

Pass/Fail Criteria:

- A student passes if marks ≥ 40
- A student fails if marks < 40

Input Format:

- Single line contains an integer representing the marks obtained by the student.

Output Format:

- Print "Pass" if the student passed the exam.
- Print "Fail" if the student failed the exam.

Sample Test Cases +

Explorer

passOrFa...

```python
marks = int(input())

if marks >= 40:
    print ("Pass")

else:
    print ("Fail")
```

Average time	Maximum time
0.007 s	0.010 s
6.71 ms	10.00 ms

3 out of 3 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 7 ms

Debug

Expected output	Actual output
45	45
Pass	Pass

Test case 2 5 ms

Test case 3 8 ms

Terminal Test cases

Algorithm:

- Start

 Input coefficients a, b, c

 Compute discriminant
$d = b^2 - 4ac$

 If $d > 0$

  - Compute
    $root1 = (-b + \sqrt{d}) / (2a)$
    $root2 = (-b - \sqrt{d}) / (2a)$
  - Print root1 and root2

 Else if $d == 0$

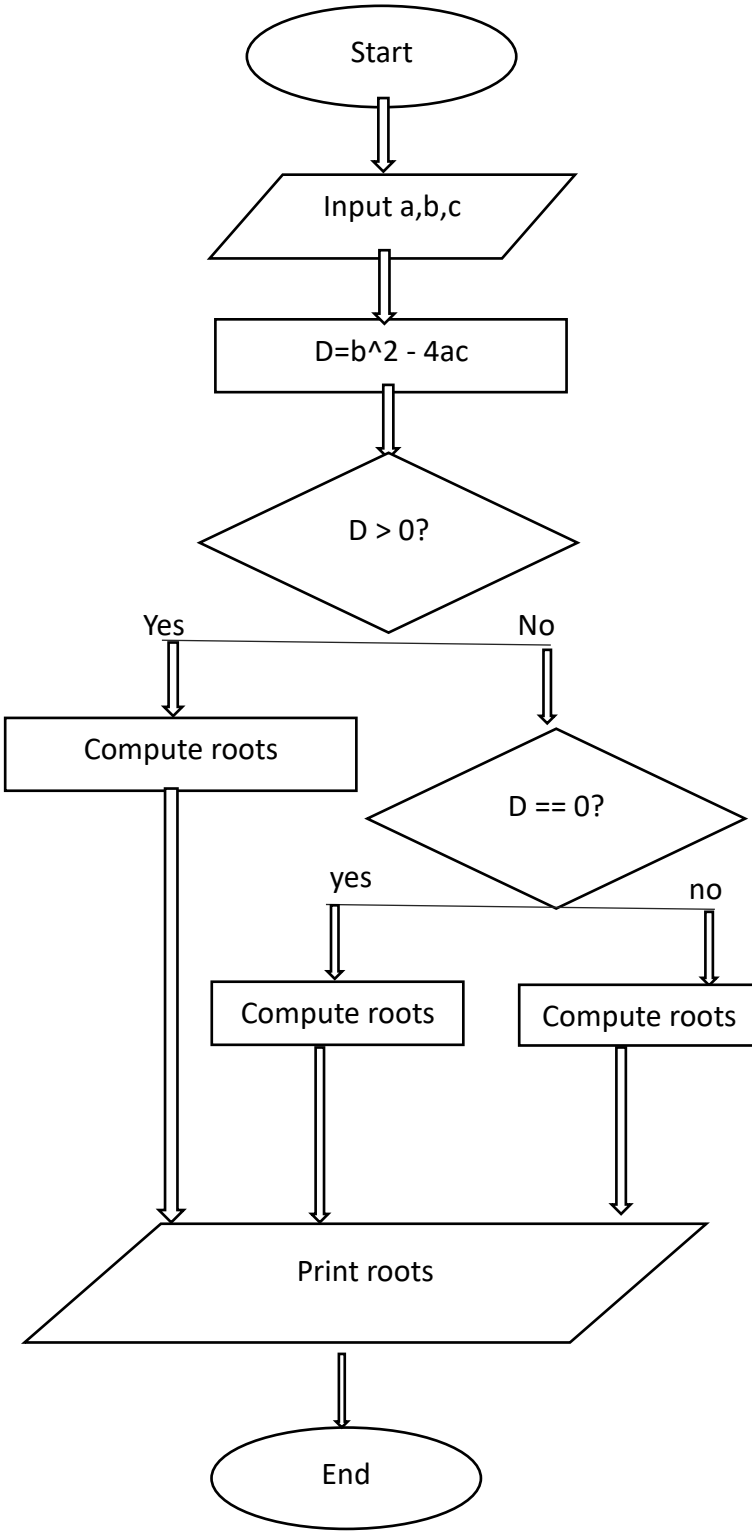
  - Compute
    $root = -b / (2a)$
  - Print root1 = root2 = root

 Else ($d < 0$)

  - Compute
    $real = -b / (2a)$
    $imag = \sqrt{(-d)} / (2a)$
  - Print
    root1 = real + imag i
    root2 = real − imag i

- End

Flowchart:



2.1.1. Roots of a Quadratic Equation

55:17

Write a program to find the roots of a quadratic equation, given its coefficients a, b, and c. Use the quadratic formula:

$$\frac{\left(-b \pm \sqrt{b^2 - 4ac}\right)}{2a}$$

The discriminant $D = b^2 - 4ac$ determines the nature of the roots:

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Input Format:

- Three space-separated integers representing the coefficients a, b, and c, respectively.

Output Format:

- If roots are real and different, print:

```
root1 = <Root1>
root2 = <Root2>
```

- If roots are the same, print:

```
root1 = root2 = <Root1>
```

- If roots are imaginary, print:

```
root1 = <RealPart>+<ImaginaryPart>i
root2 = <RealPart>-<ImaginaryPart>i
```

- All values should be formatted to two decimal places.

Sample Test Cases +

Explorer

quadratic... Submit

```python
import math
a, b, c = map(int, input().split())
d = b*b - 4*a*c

if d > 0:
    r1 = (-b + math.sqrt(d)) / (2*a)
    r2 = (-b - math.sqrt(d)) / (2*a)
    print(f"root1 = {r1:.2f}")
    print(f"root2 = {r2:.2f}")

elif d == 0:
    r = -b / (2*a)
    print(f"root1 = root2 = {r:.2f}")
else:
    real = -b / (2*a)
    imag = math.sqrt(-d) / (2*a)
    print(f"root1 = {real:.2f}+{imag:.2f}i")
    print(f"root2 = {real:.2f}-{imag:.2f}i")
```

Debugger

Average time	Maximum time
0.009 s	0.017 s
9.33 ms	17.00 ms

3 out of 3 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 17 ms — Debug

Expected output	Actual output
1 -5 6	1 -5 6
root1 = 3.00	root1 = 3.00
root2 = 2.00	root2 = 2.00

Test case 2 7 ms

Terminal Test cases